

Decimal Codes

1 Binary-Coded Decimal (BCD)

Definition

- A **weighted** code where each decimal digit (0–9) is represented by its binary equivalent using **4 bits**.
- Also known as **8421 code** due to the positional weights: 8, 4, 2, 1.

Table

Decimal	BCD (8421)
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Features

- Easy decimal-to-binary conversion.
- Unused codes: 1010 to 1111 (invalid in standard BCD).
- Used in calculators, digital clocks, etc.

2 ASCII Code (American Standard Code for Information Interchange)

Definition

- ASCII is a **7-bit** character encoding scheme that represents **text characters** (letters, digits, symbols) in computers.
- Decimal digits (0–9) are encoded in a specific 7-bit binary form.

Table for Decimal Digits

Decimal Digit	ASCII (Decimal)	ASCII (Binary)
0	48	0110000
1	49	0110001
2	50	0110010
3	51	0110011
4	52	0110100
5	53	0110101
6	54	0110110
7	55	0110111
8	56	0111000
9	57	0111001

Applications

- Widely used in programming, file formats, networking, and text display systems.

3 Excess-3 (XS-3) Code

Definition

- A **non-weighted, self-complementary** code.
- Each decimal digit is represented by its 4-bit binary equivalent **plus 3 (0011)**.

Table

Decimal	Excess-3
0	0011
1	0100
2	0101
3	0110
4	0111
5	1000
6	1001
7	1010
8	1011
9	1100

Features

- Useful in arithmetic operations and error detection.
- Self-complementary: 9's complement of a digit can be obtained by bitwise NOT.

4 Two-Out-of-Five Code (2-of-5)

Definition

- A **weighted code** where exactly **two bits are ‘1’** and the rest are ‘0’ in a 5-bit group.
- Used in barcode systems and optical character recognition.

Weights: 7, 4, 2, 1, 0 (common scheme)

Decimal	2-of-5 Code
0	11000
1	00011
2	00101
3	00110
4	01001
5	01010
6	01100
7	10001
8	10010
9	10100

Features

- Good for error detection.
- Each code word has exactly two 1s → easy to validate.

5 Summary Table

Code Type	Bit Length	Type	Special Feature	Application Area
BCD (8421)	4	Weighted	Simple decimal conversion	Digital clocks, calculators
ASCII	7	Character Code	Standard text encoding	Programming, text display
Excess-3	4	Non-weighted	Self-complementary	Arithmetic and logic ops
Two-out-of-5	5	Weighted	Fixed weight, error check	Barcode, OCR systems